

Cloud Computing

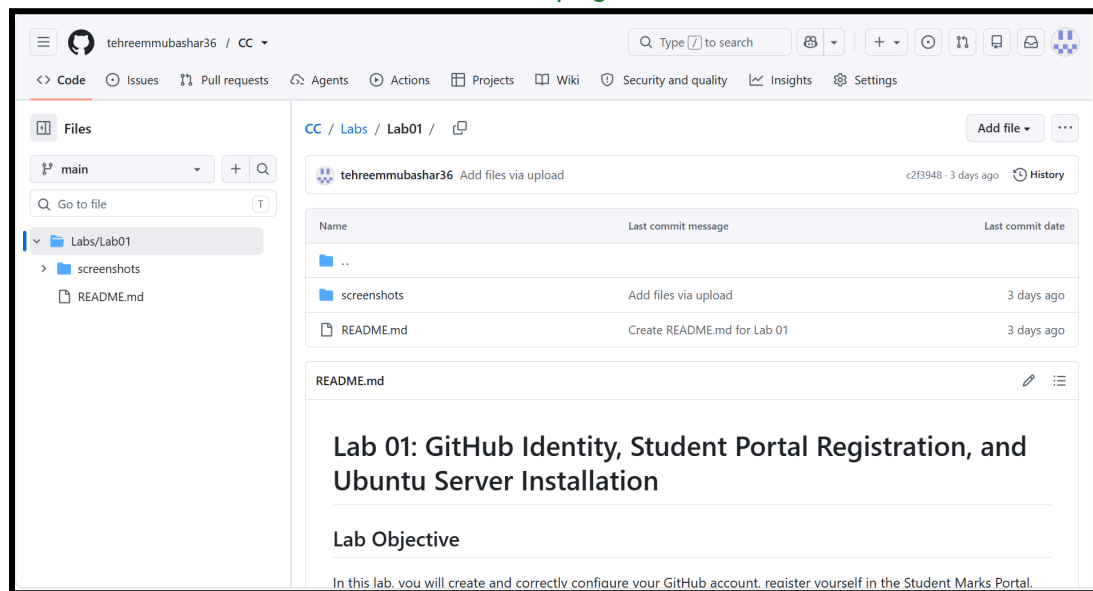
Lab 01

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

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<i>Reg #</i>	FA24B1-SE-036
<i>Semester</i>	5th
<i>Course Code</i>	SE-665
<i>Github username</i>	tehreemmubashar36

Getting Started

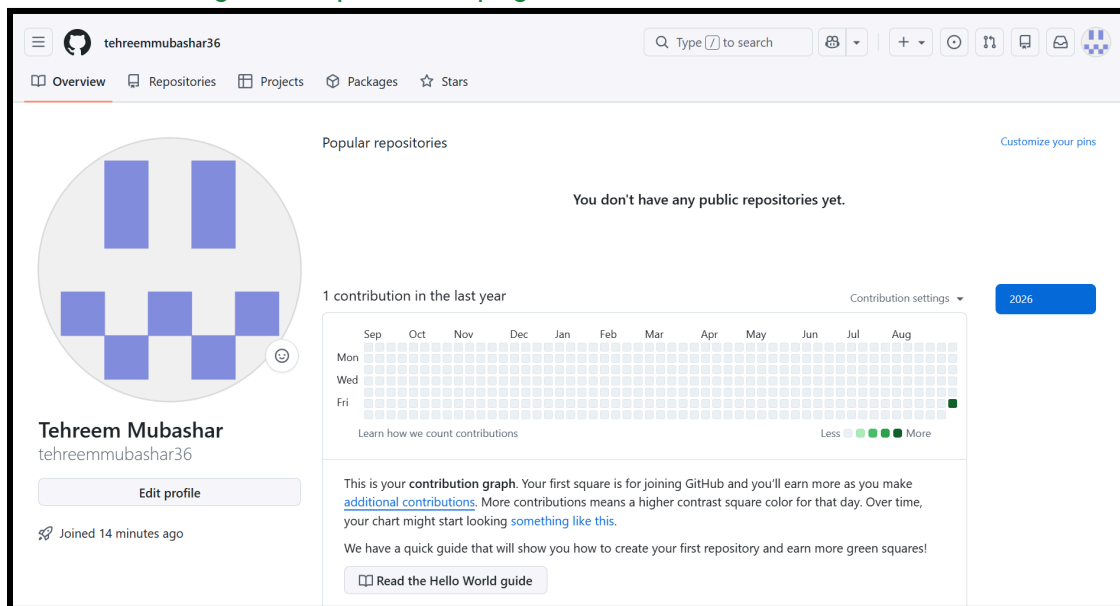
- Screenshot: [lab01_folder_structure.png](#)



Description: This screenshot shows the created Labs/Lab01/screenshots/ directory structure inside the CC repository.

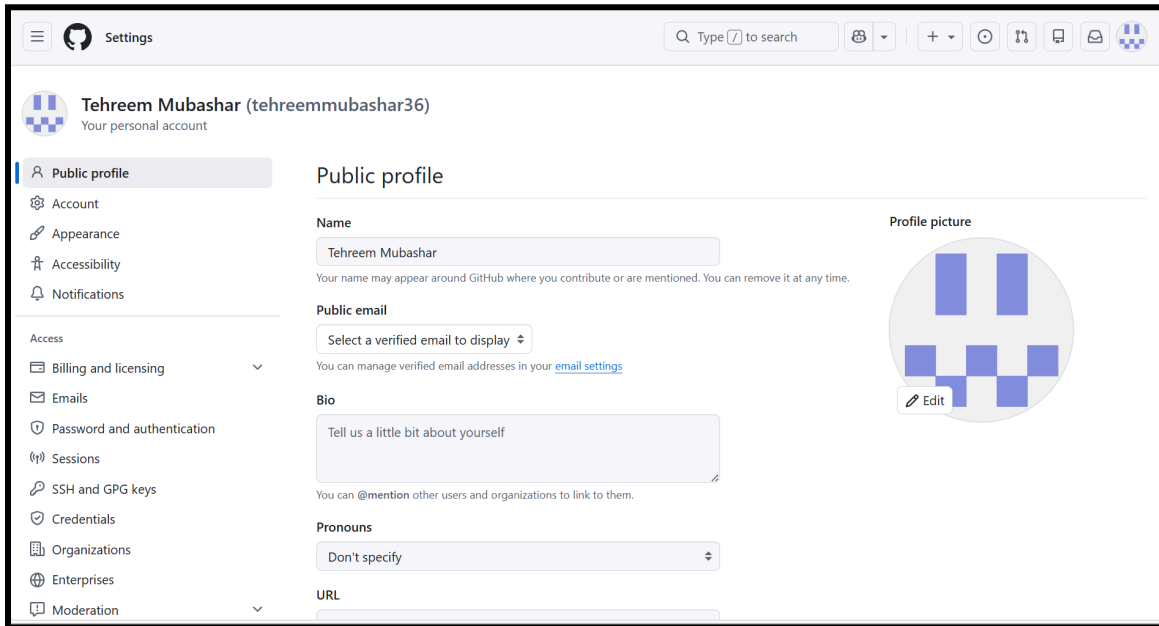
Task 1: Create and Configure Your GitHub Account

- Screenshot: [github_profile.png](#)



Description: This task shows the creation and configuration of the GitHub account using the student's actual identity and a professional username.

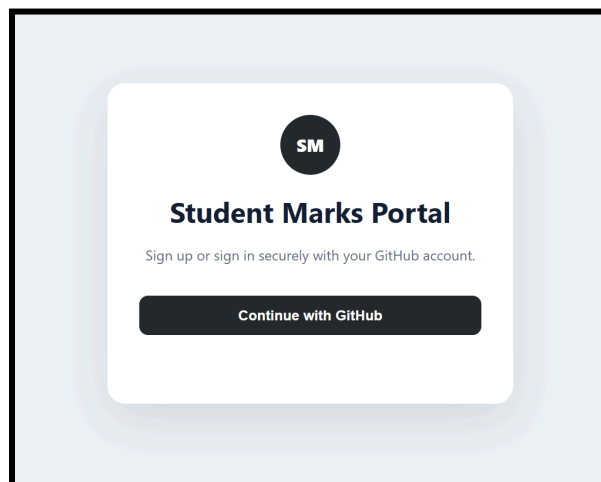
- Screenshot: [github_actual_name.png](#)



Description: This screenshot shows the student's complete actual name entered in the GitHub Name field.

Task 2: Register in the Student Marks Portal

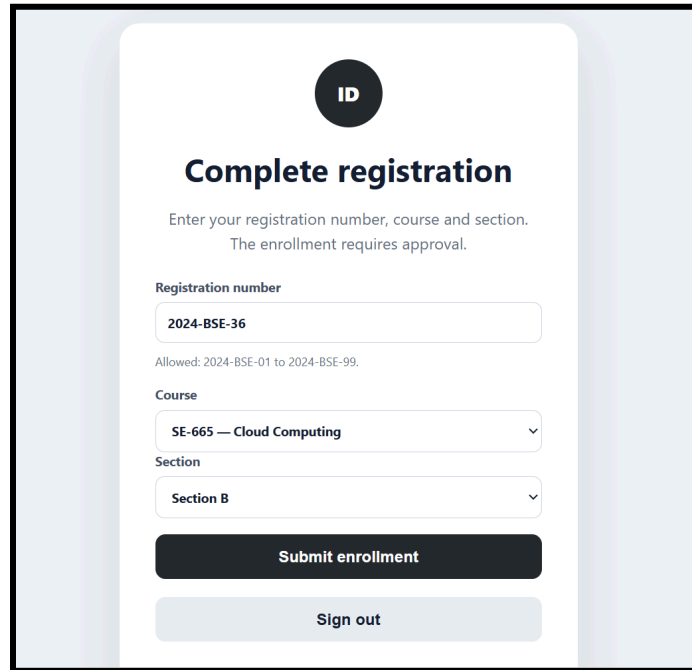
- Screenshot: [portal_login_page.png](#)



Description:

This task shows registration in the Student Marks Portal using the same GitHub account created in Task 1.

- Screenshot: [portal_github_profile.png](#)

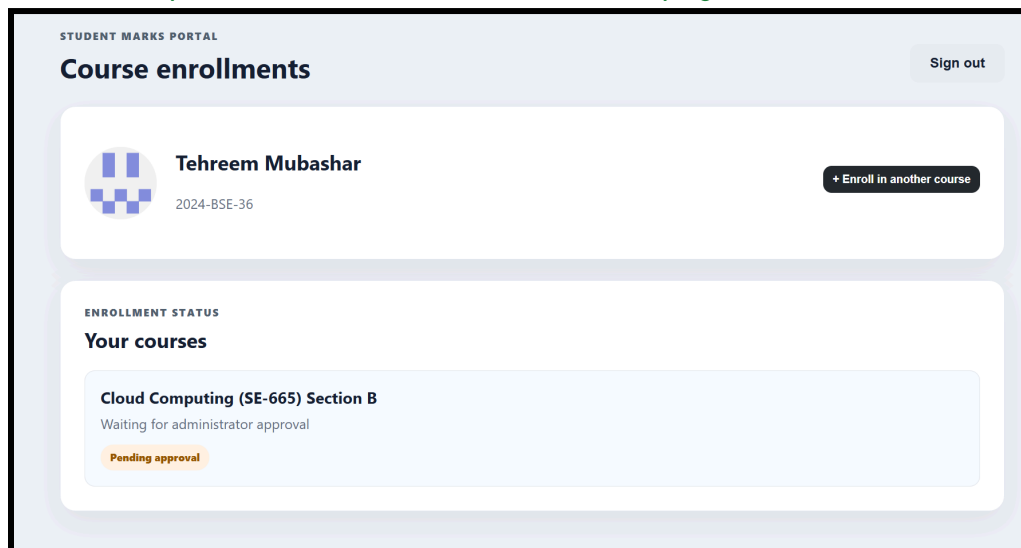


The screenshot shows a registration form titled "Complete registration". At the top, there is a circular profile icon with the letters "ID". Below the title, a message states: "Enter your registration number, course and section. The enrollment requires approval." The form contains three input fields: "Registration number" with the value "2024-BSE-36", "Course" with the value "SE-665 — Cloud Computing", and "Section" with the value "Section B". Below these fields is a dark blue button labeled "Submit enrollment" and a light blue button labeled "Sign out". A small note below the registration number field says "Allowed: 2024-BSE-01 to 2024-BSE-99."

Description:

This screenshot shows the student's GitHub profile information displayed inside the Student Marks Portal.

- Screenshot: [portal_enrollment_submitted.png](#)



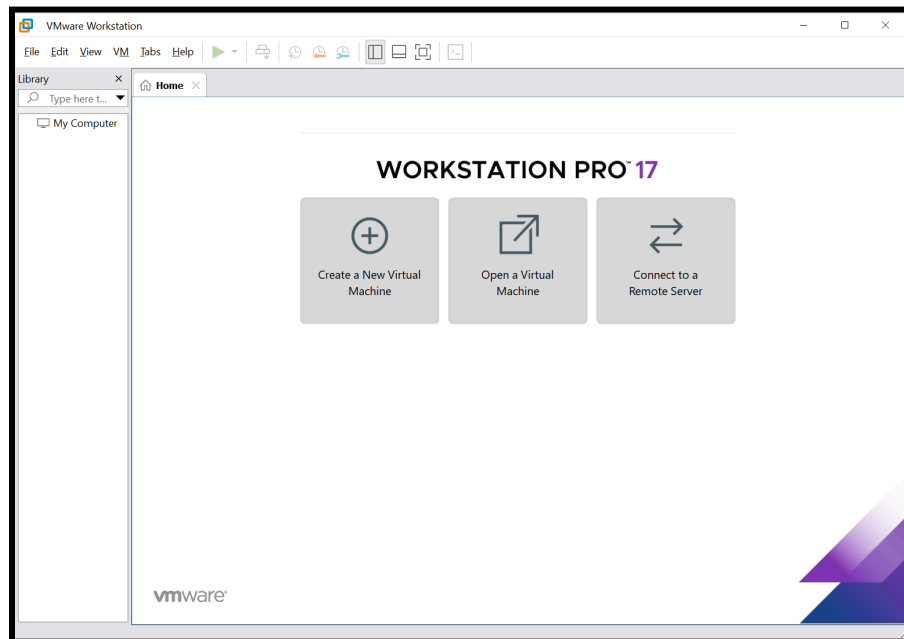
The screenshot shows the "STUDENT MARKS PORTAL" header with a "Sign out" button. The main section is titled "Course enrollments". It features a student profile for "Tehreem Mubashar" with the ID "2024-BSE-36" and a button to "+ Enroll in another course". Below this, the "ENROLLMENT STATUS" section is titled "Your courses". It lists "Cloud Computing (SE-665) Section B" with the status "Waiting for administrator approval" and a yellow "Pending approval" tag.

Description:

This screenshot shows the enrollment submitted or pending for approval in the Student Marks Portal.

Task 3: Download VMware Workstation Pro and Ubuntu Server

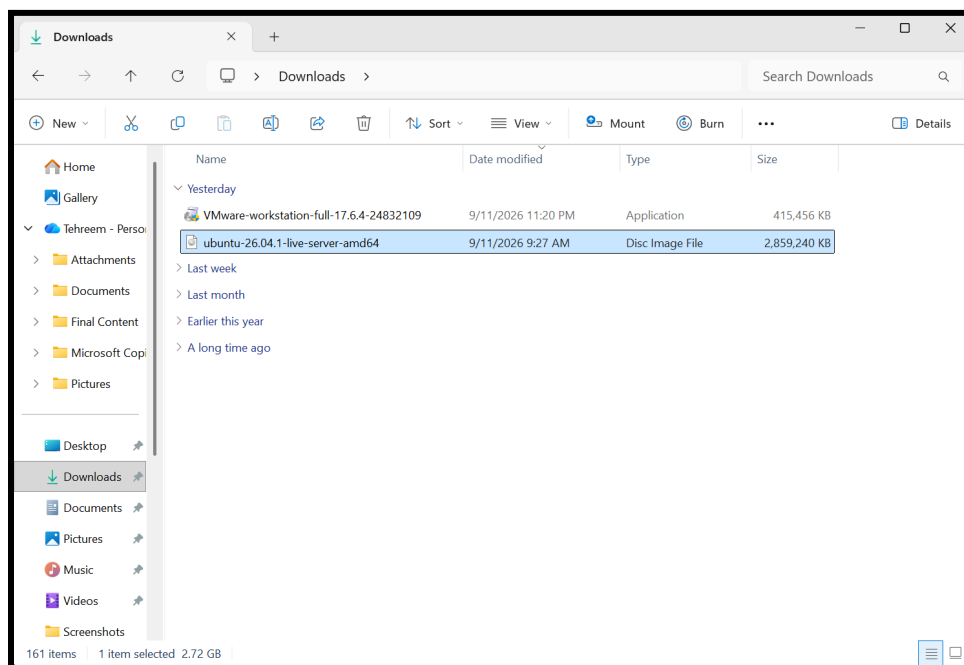
- Screenshot: [vmware_workstation.png](#)



Description:

This screenshot shows VMware Workstation Pro running successfully on the Windows computer.

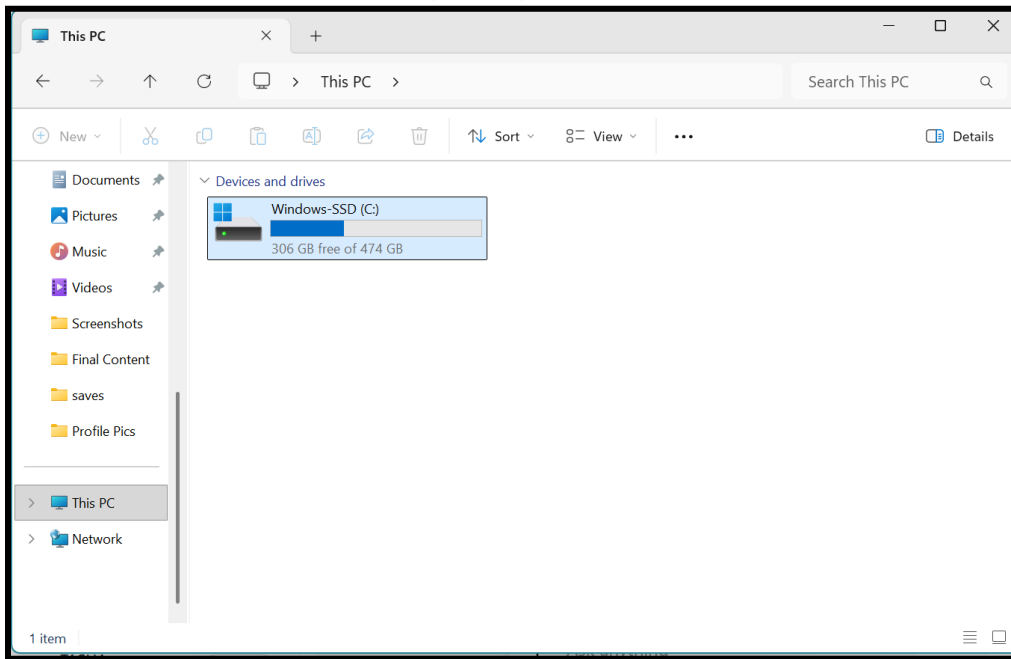
- Screenshot: [ubuntu_server_iso.png](#)



Description:

This screenshot shows the downloaded Ubuntu Server LTS ISO file and its filename.

- Screenshot: [available_storage.png](#)

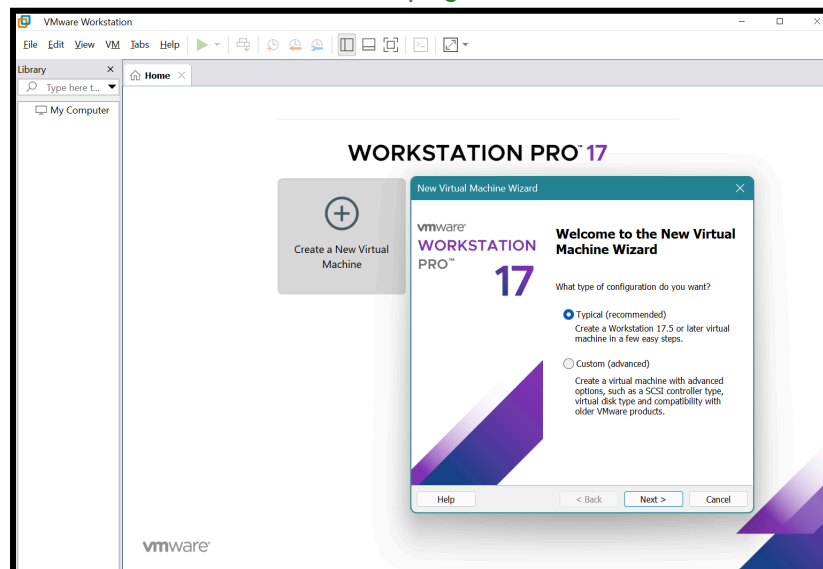


Description:

This screenshot shows that at least 20 GB of free storage is available for the Ubuntu Server virtual machine.

Task 4: Create the Ubuntu Server Virtual Machine

- Screenshot: [vm_creation_wizard.png](#)



Description:

This screenshot shows the VMware New Virtual Machine Wizard used to create the Ubuntu Server virtual machine.

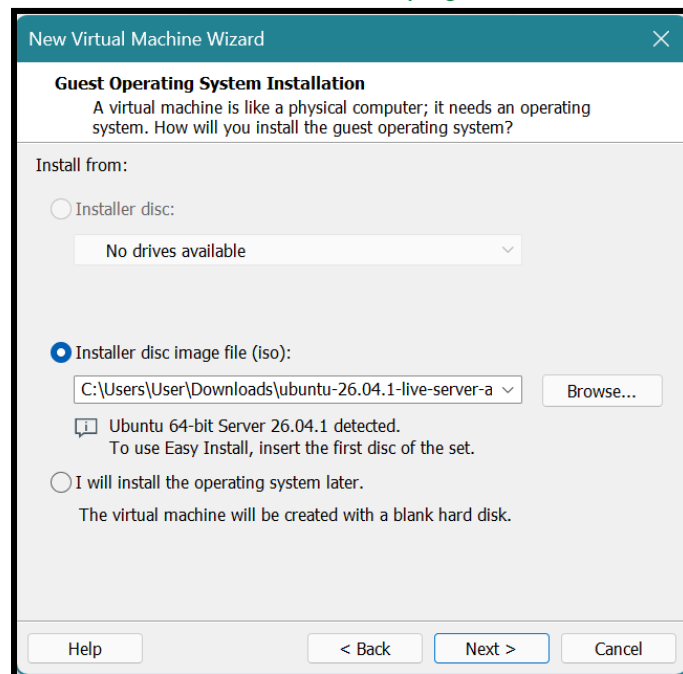
- Screenshot: [vm_typical_configuration.png](#)



Description:

This screenshot shows **Typical (recommended)** selected as the virtual machine configuration.

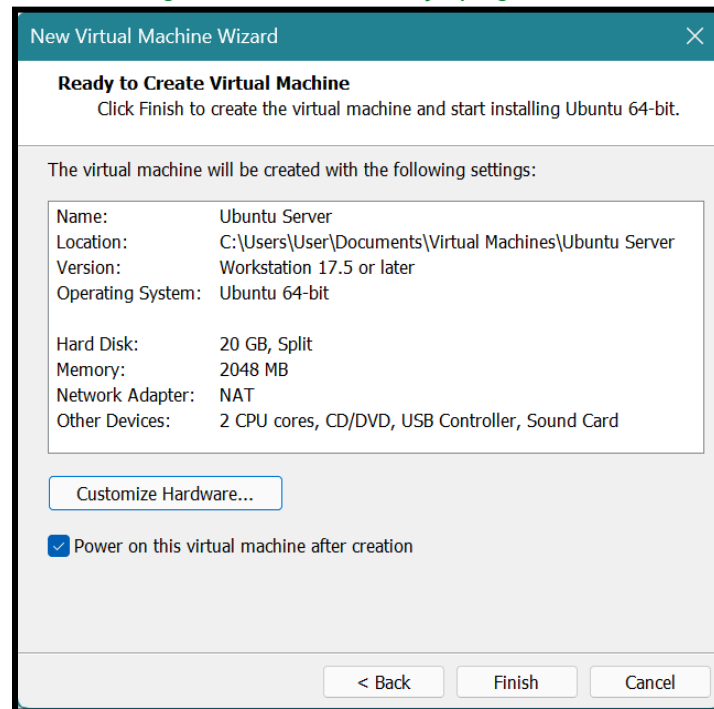
- Screenshot: [vm_ubuntu_iso_selected.png](#)



Description:

This screenshot shows the Ubuntu Server ISO selected as the installer disc image.

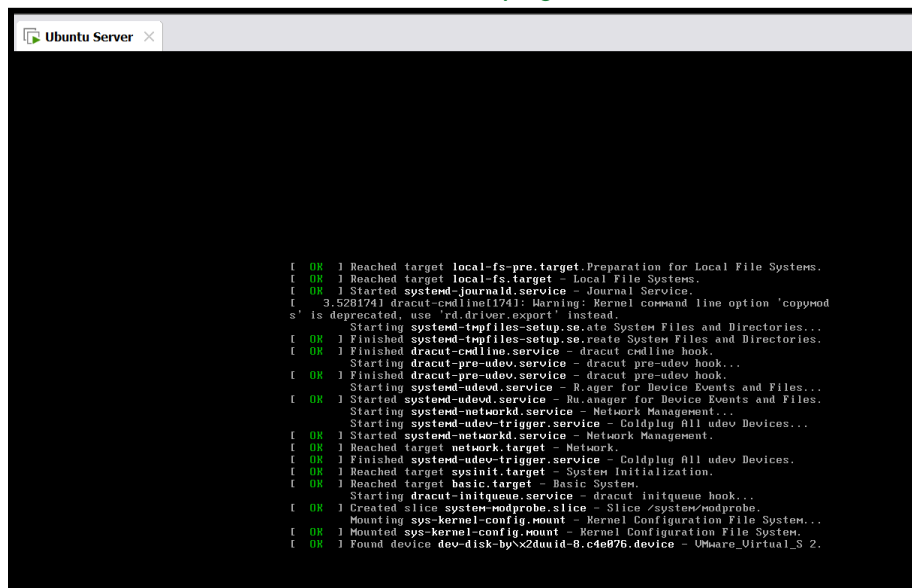
- Screenshot: [vm_configuration_summary.png](#)



Description:

This screenshot shows the final virtual machine configuration with the required virtual disk allocation.

- Screenshot: [ubuntu_installer_boot.png](#)

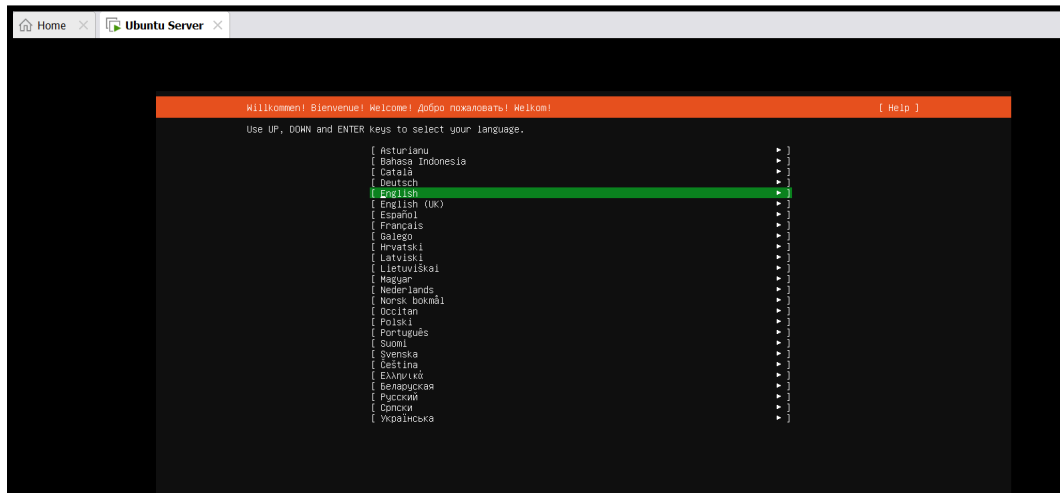


Description:

This screenshot shows the Ubuntu Server installer boot screen after powering on the virtual machine.

Task 5: Choose the Installation Language

- Screenshot: [ubuntu_language.png](#)

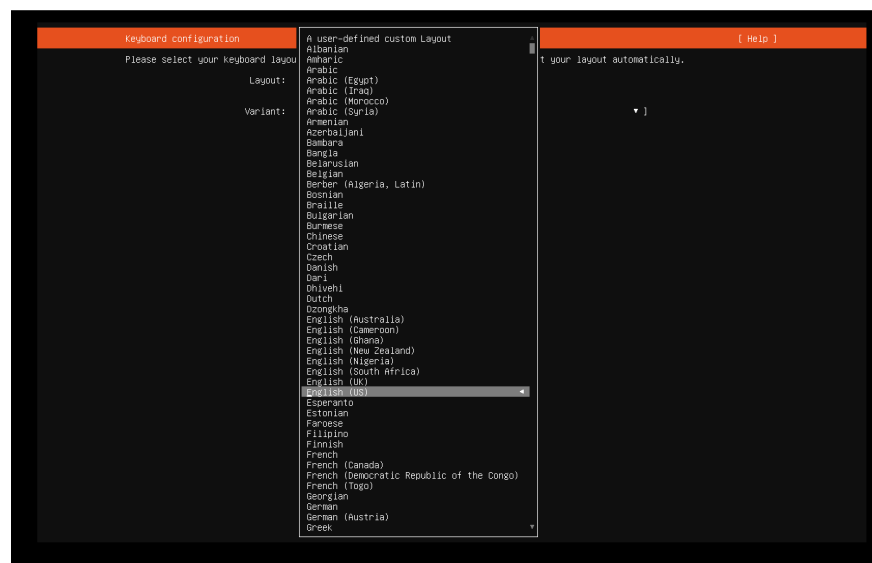


Description:

This screenshot shows the required installation language selected in the Ubuntu Server installer.

Task 6: Configure the Keyboard Layout

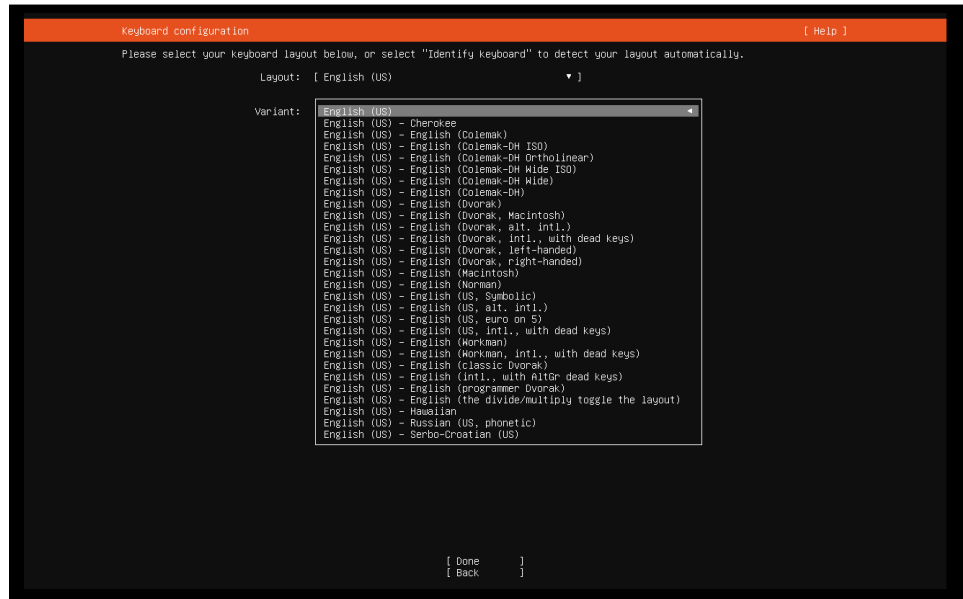
- Screenshot: [ubuntu_keyboard_layout.png](#)



Description:

This screenshot shows the keyboard layout selected to match the computer's keyboard.

- Screenshot: [ubuntu_keyboard_variant.png](#)

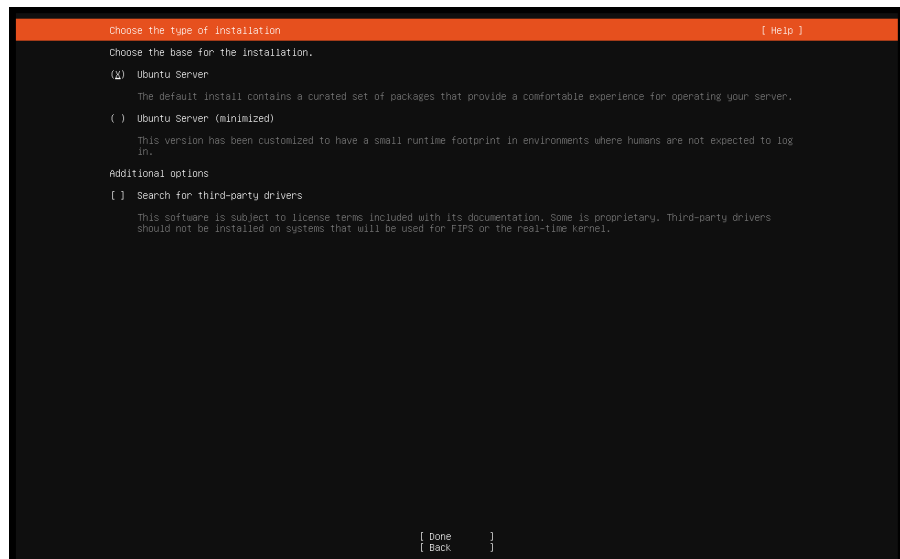


Description:

This screenshot shows the appropriate keyboard variant selected for the installation.

Task 7: Choose the Ubuntu Server Installation Type

- Screenshot: [ubuntu_installation_type.png](#)

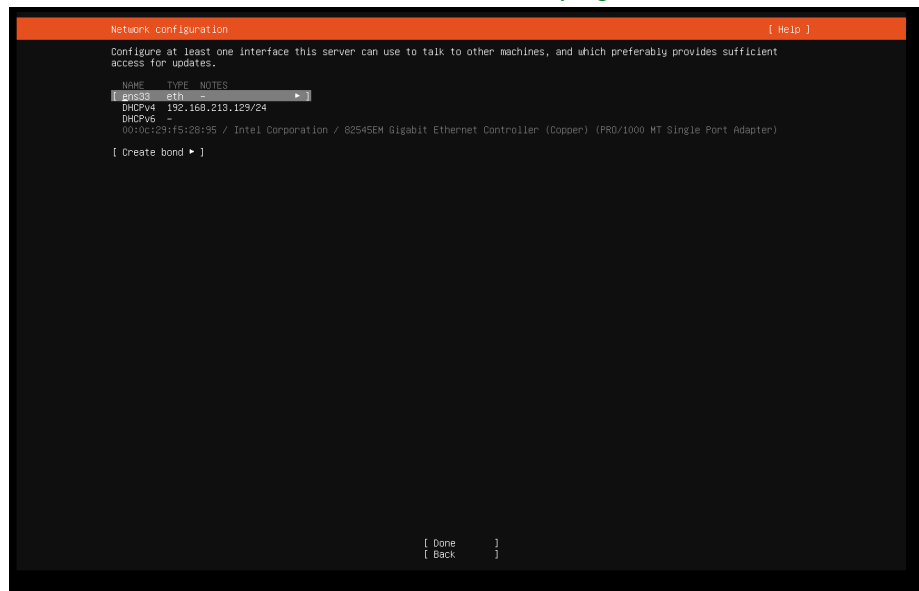


Description:

This screenshot shows the normal **Ubuntu Server** installation option selected.

Task 8: Configure the Network

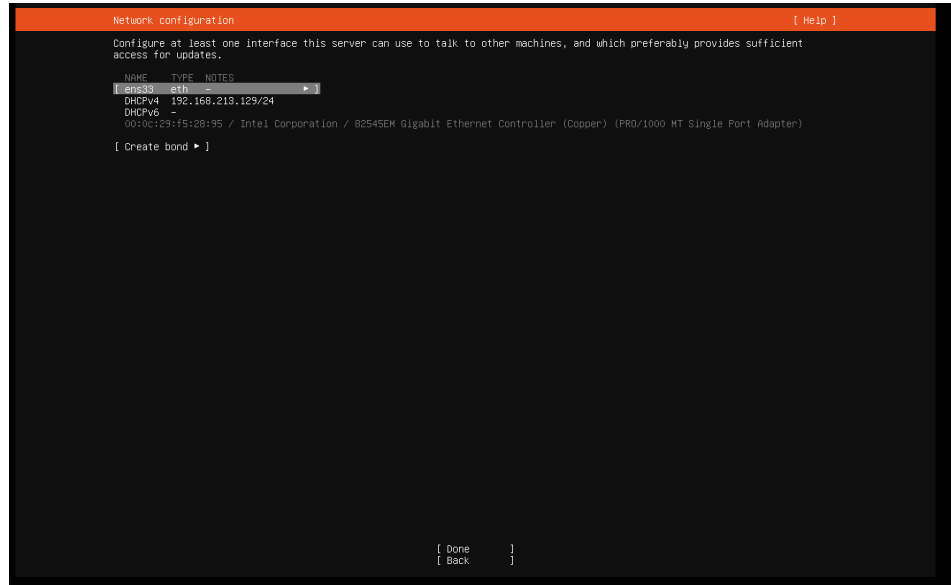
- Screenshot: [ubuntu_network_interface.png](#)



Description:

This screenshot shows the network interface detected by the Ubuntu Server installer.

- Screenshot: [ubuntu_installer_network.png](#)

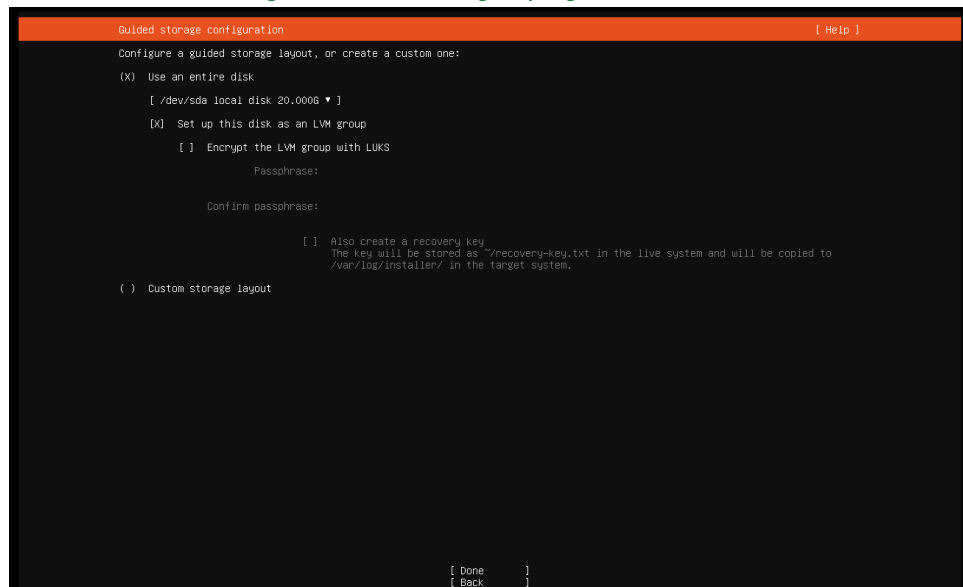


Description:

This screenshot shows the IP/network address automatically assigned to the Ubuntu Server through DHCP.

Task 9: Configure Storage

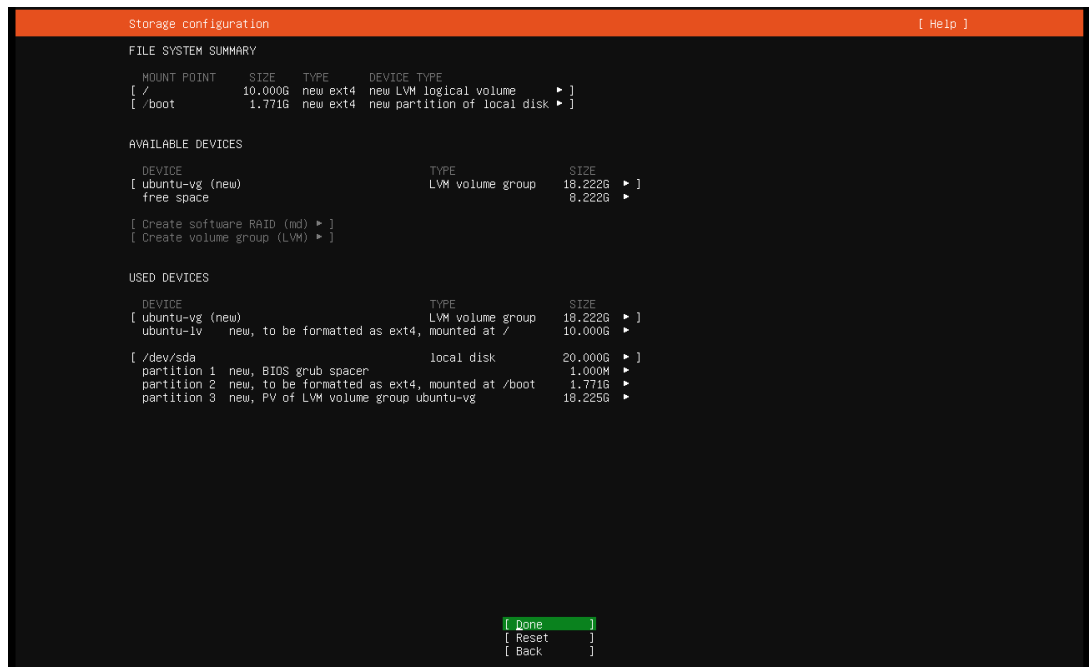
- Screenshot: [ubuntu_guided_storage.png](#)



Description:

This screenshot shows the guided storage option selected for the VMware virtual disk.

- Screenshot: [ubuntu_storage_configuration.png](#)

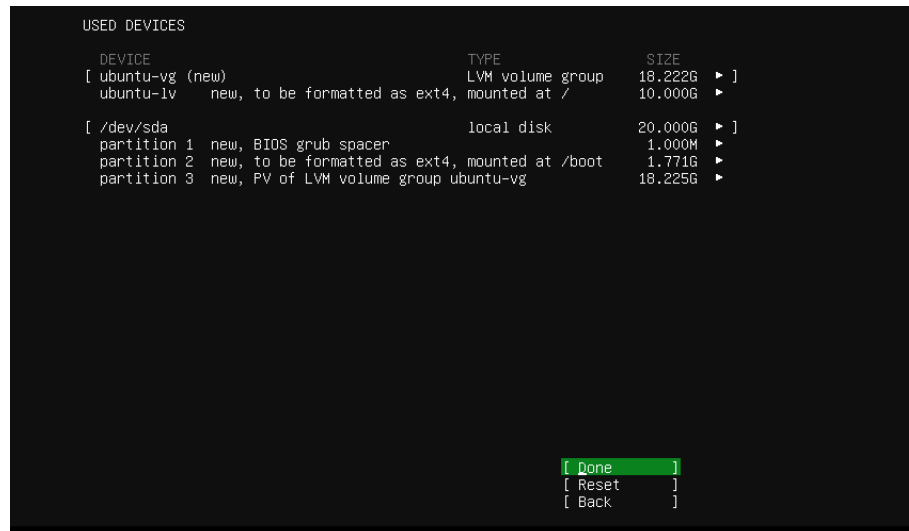


Description:

This screenshot shows the proposed storage configuration before continuing with the installation.

Task 10: Select the Installation Disk

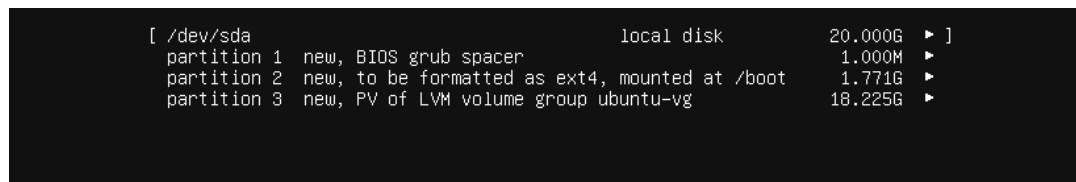
- Screenshot: [ubuntu_selected_disk.png](#)



Description:

This screenshot shows the virtual disk selected for installing Ubuntu Server.

- Screenshot: [ubuntu_disk_details.png](#)

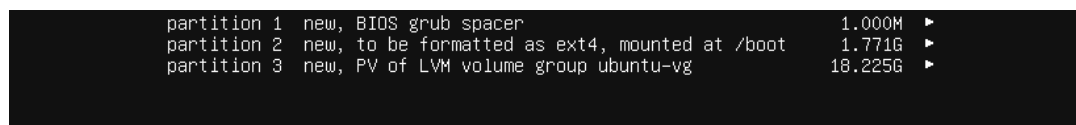


Description:

This screenshot shows the selected VMware virtual disk name and its size.

Task 11: Review the Partitions

Screenshot: [ubuntu_partition_layout.png](#)

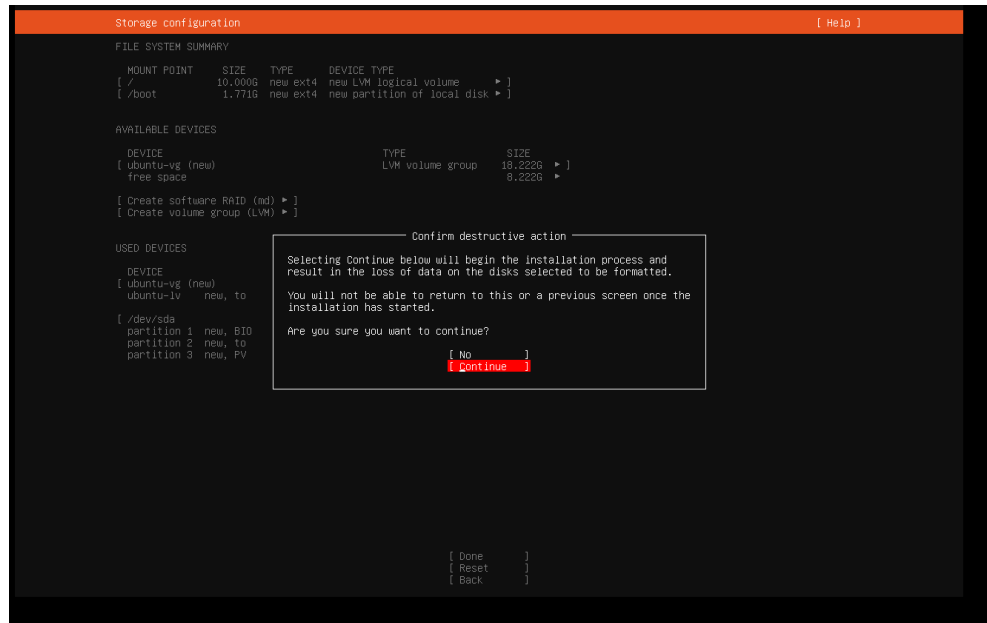


Description:

This screenshot shows the partition layout proposed by the Ubuntu Server installer for the VMware virtual disk.

Task 12: Confirm the Storage Changes

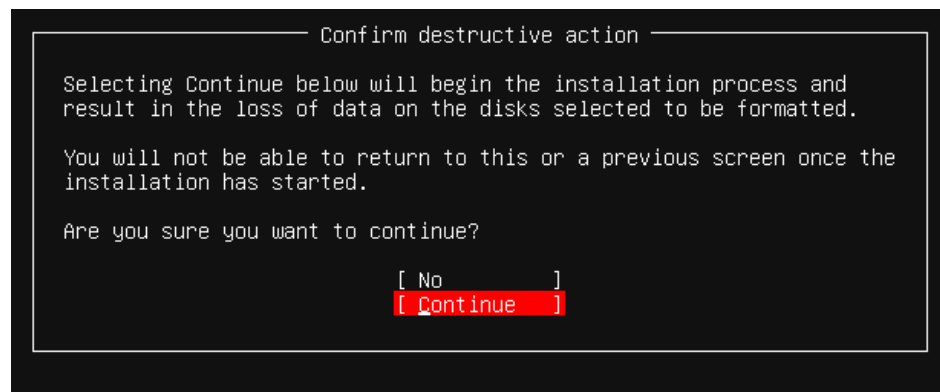
- Screenshot: [ubuntu_storage_warning.png](#)



Description:

This screenshot shows the final warning before applying the storage changes to the selected VMware virtual disk.

- Screenshot: [ubuntu_storage_confirmed.png](#)



Description:

This screenshot shows the confirmation to apply the proposed storage changes.

Task 13: Configure the Ubuntu Profile

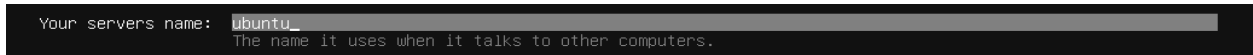
- Screenshot: [ubuntu_actual_name.png](#)



Description:

This screenshot shows the student's complete actual name entered in the **Your name** field.

- Screenshot: [ubuntu_server_name.png](#)



Description:

This screenshot shows **ubuntu** entered as the Ubuntu Server name.

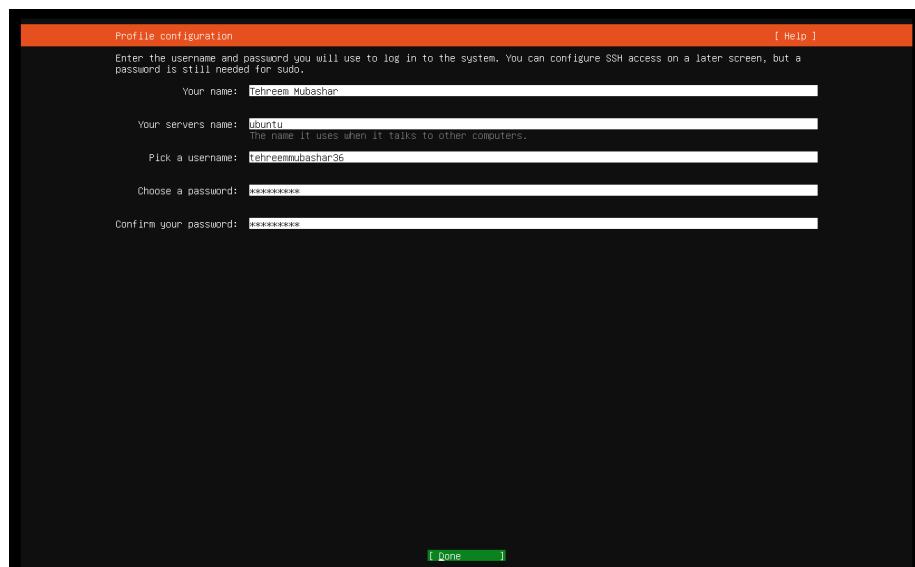
- Screenshot: [ubuntu_username.png](#)



Description:

This screenshot shows the exact GitHub username entered as the Ubuntu username.

- Screenshot: [ubuntu_profile_setup.png](#)

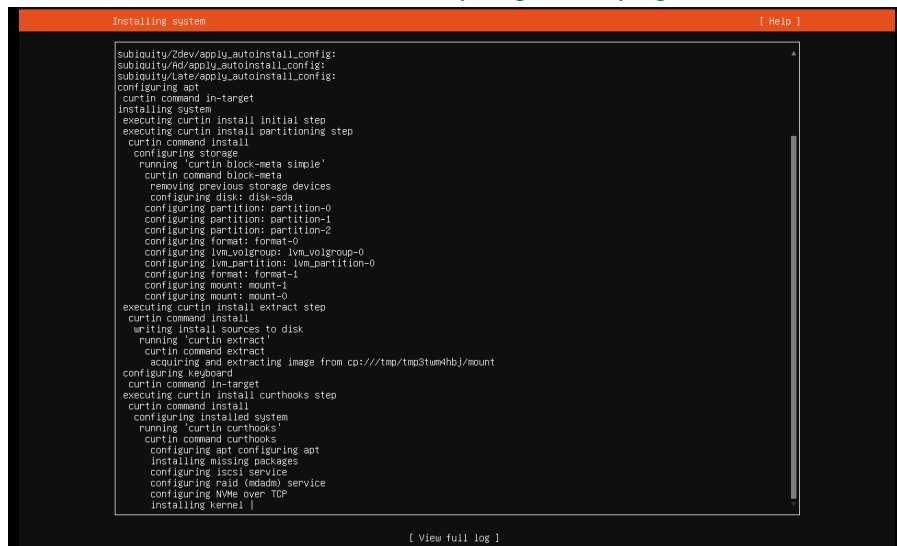


Description:

This screenshot shows the completed Ubuntu profile information before continuing with the installation.

Task 14: Complete the Software Installation

- Screenshot: [ubuntu_installation_progress.png](#)



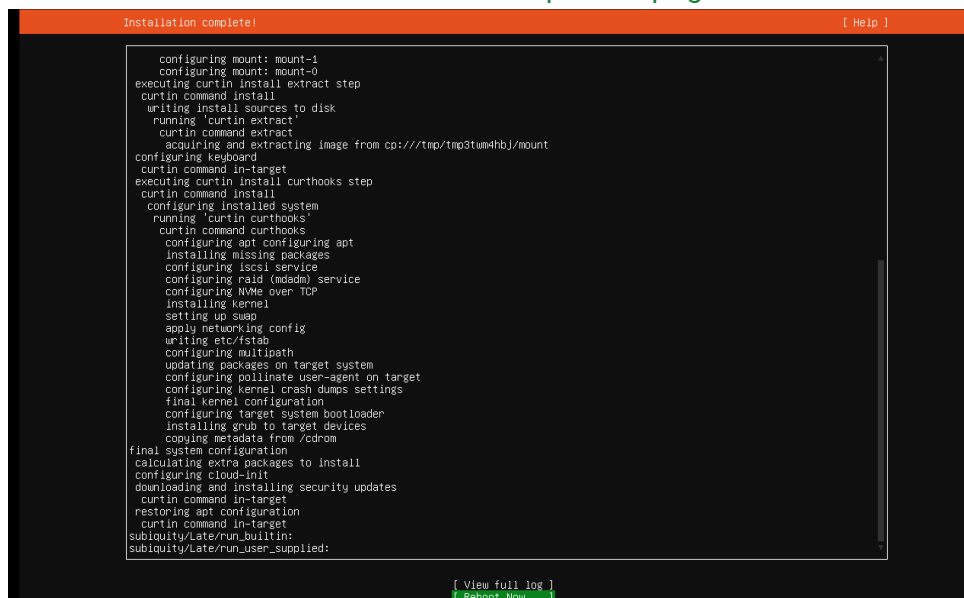
The screenshot shows a terminal window titled "Installing system" with a red header bar. The terminal displays the progress of the Ubuntu Server installation, including steps like configuring apt, installing system, executing curtin install initial step, executing curtin install partitioning step, configuring storage, running 'curtin block-meta simple', removing previous storage devices, configuring disk: disk-sda, configuring partitions partition-0, partition-1, partition-2, configuring format: format-0, configuring lvm.volgroups lvm.volgroup-0, configuring lvm.partition: lvm.partition-0, configuring format: format-1, configuring mount: mount-1, configuring mount: mount-0, executing curtin install extract step, curtin command install, writing install sources to disk, running 'curtin extract', curtin command extract, acquiring and extracting image from cp:///tmp/tpm3tmm4hbj/mount, configuring keyboard, curtin command in-target, executing curtin install curthooks step, curtin command install, configuring installed system, running 'curtin curthooks', curtin command curthooks, configuring apt, configuring apt, installing missing packages, configuring iscsi service, configuring raid (mdadm) service, configuring iWoe over TCP, installing kernel, and installing kernel. At the bottom, there is a link to "View full log".

Description:

This screenshot shows Ubuntu Server actively installing and configuring the required system software.

Task 15: Reboot and Verify the Ubuntu Identity

- Screenshot: [ubuntu_installation_complete.png](#)



The screenshot shows a terminal window titled "Installation complete!" with a red header bar. The terminal displays the final steps of the Ubuntu Server installation, including configuring mount: mount-1, configuring mount: mount-0, executing curtin install extract step, curtin command install, writing install sources to disk, running 'curtin extract', curtin command extract, acquiring and extracting image from cp:///tmp/tpm3tmm4hbj/mount, configuring keyboard, curtin command in-target, executing curtin install curthooks step, curtin command install, configuring installed system, running 'curtin curthooks', curtin command curthooks, configuring apt, configuring apt, installing missing packages, configuring iscsi service, configuring raid (mdadm) service, configuring iWoe over TCP, installing kernel, setting up swap, apply networking config, writing etc/iscsi, configuring multipath, updating packages on target system, configuring pollinate user-agent on target, configuring kernel crash dumps settings, final kernel configuration, configuring target system bootloader, installing grub to target devices, copying metadata from /cdrom, final system configuration, calculating extra packages to install, configuring cloud-init, downloading and installing security updates, curtin command in-target, restoring apt configuration, curtin command in-target, subiquity/late/run_builtins, and subiquity/late/run_user_supplied. At the bottom, there are links to "View full log" and "Reboot Now".

Description:

This screenshot shows that the Ubuntu Server installation has completed and the system is ready to reboot.

- Screenshot: [ubuntu_login_screen.png](#)

```
Ubuntu 26.04.1 LTS ubuntu tty1
ubuntu login:
ubuntu login: tehreemmubashar36
Password:
```

Description:

This screenshot shows the Ubuntu Server login screen after the virtual machine has rebooted.

- Screenshot: [ubuntu_terminal_login.png](#)

```
Ubuntu 26.04.1 LTS ubuntu tty1
ubuntu login: tehreemmubashar36
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat Sep 12 11:59:19 AM UTC 2026

System load:  0.24          Processes:      251
Usage of /:   50.6% of 9.75GB Users logged in:  0
Memory usage: 21%          IPv4 address for ens33: 192.168.213.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

7 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

tehreemmubashar36@ubuntu:~$
```

Description:

This screenshot shows the successful login to the installed Ubuntu Server terminal.

- Screenshot: [ubuntu_identity_verified.png](#)

```
tehreemmubashar36@ubuntu:~$
```

Description:

This screenshot verifies that the Ubuntu terminal prompt follows the required `<github-username>@ubuntu:~$` format.

Task 16: Find the Ubuntu Server IP Address

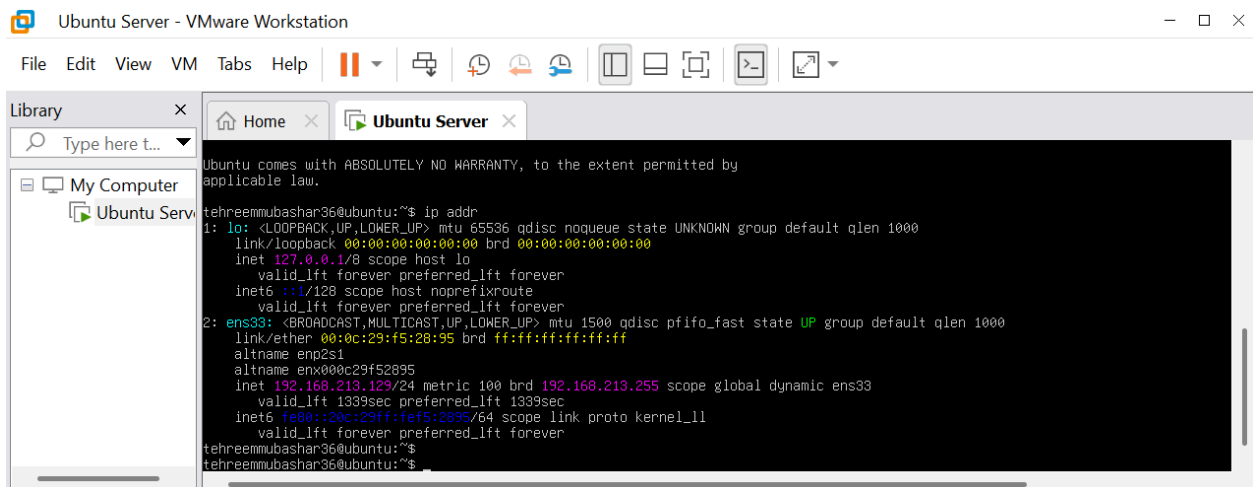
- Screenshot: [ubuntu_ip_addr_command.png](#)

```
tehreemmubashar36@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:f5:28:95 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29f52895
    inet 192.168.213.129/24 metric 100 brd 192.168.213.255 scope global dynamic ens33
        valid_lft 1339sec preferred_lft 1339sec
    inet6 fe80::20c:29ff:fe5:2895/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
tehreemmubashar36@ubuntu:~$ _
```

Description:

This screenshot shows the `ip addr` command and its output used to identify the Ubuntu Server network information.

- Screenshot: [ubuntu_ip_address.png](#)



Description:

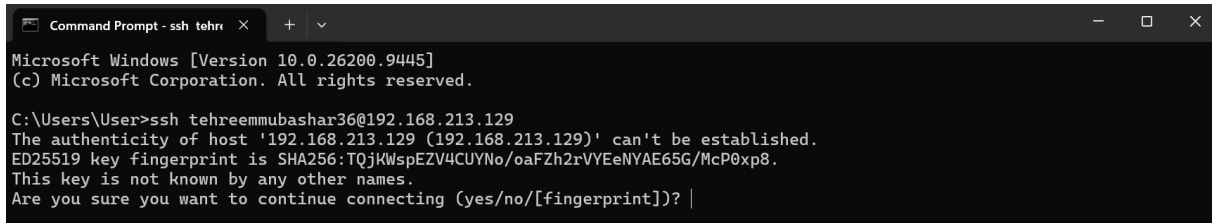
This screenshot clearly shows the correct IPv4 address of the active Ubuntu Server network interface.

For this screenshot, the correct IP you found was:

192.168.213.129

Task 17: Connect to Ubuntu Server from Windows

- Screenshot: [windows_ssh_command.png](#)



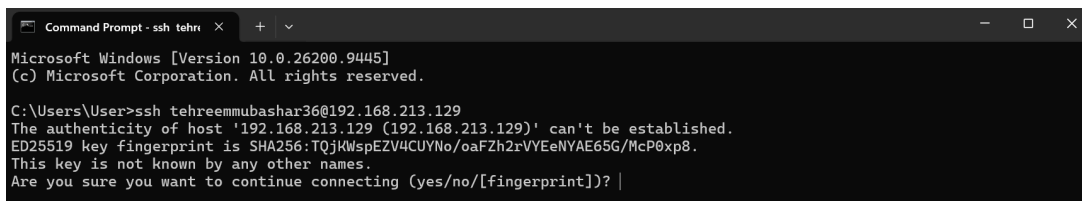
```
Command Prompt - ssh tehre X + v
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh tehreemubashar36@192.168.213.129
The authenticity of host '192.168.213.129 (192.168.213.129)' can't be established.
ED25519 key fingerprint is SHA256:TQjKWspEZV4CUYN0/oaFZh2rVYEeNYAE65G/McP0xp8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

Description:

This screenshot shows the SSH command used from Windows to connect to the Ubuntu Server.

- Screenshot: [windows_ssh_fingerprint.png](#)



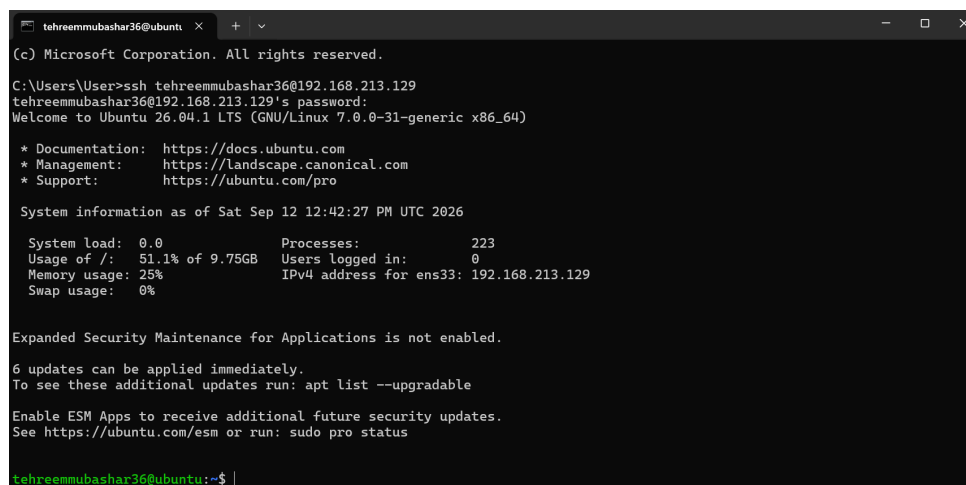
```
Command Prompt - ssh tehre X + v
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh tehreemubashar36@192.168.213.129
The authenticity of host '192.168.213.129 (192.168.213.129)' can't be established.
ED25519 key fingerprint is SHA256:TQjKWspEZV4CUYN0/oaFZh2rVYEeNYAE65G/McP0xp8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

Description:

This screenshot shows the Ubuntu Server SSH fingerprint being accepted by the Windows client.

- Screenshot: [windows_ssh_login.png](#)



```
tehreemubashar36@ubuntu X + v
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh tehreemubashar36@192.168.213.129
tehreemubashar36@192.168.213.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Sep 12 12:42:27 PM UTC 2026

System load:  0.0          Processes:    223
Usage of /:   51.1% of 9.75GB Users logged in:  0
Memory usage: 25%         IPv4 address for ens33: 192.168.213.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.
6 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

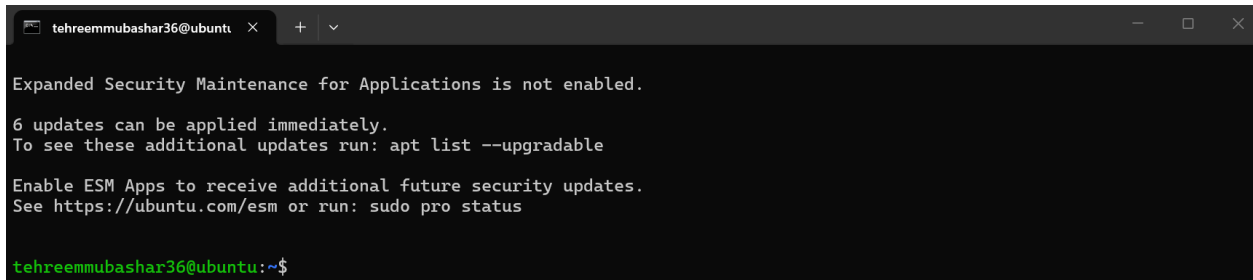
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

tehreemubashar36@ubuntu:~$ |
```

Description:

This screenshot shows the successful SSH session from Windows to the Ubuntu Server.

- Screenshot: `windows_ssh_identity.png`



```
tehreemmubashar36@ubuntu x + v
Expanded Security Maintenance for Applications is not enabled.
6 updates can be applied immediately.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
tehreemmubashar36@ubuntu:~$
```

Description:

This screenshot verifies that the remote SSH terminal prompt displays the required `<github-username>@ubuntu` identity.